

Advanced Topics in Computer Graphics I

Monte Carlo Integration



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May 13, 2026



The Rendering Equation is defined as

$$L_o(\mathbf{x}, \Omega_o) = L_e(\mathbf{x}, \Omega_o) + \int_{\mathcal{H}_i} f_{BRDF}(\Omega_i, \mathbf{x}, \Omega_o) L_i(\mathbf{x}, \Omega_i) \cos(\theta_i) d\omega_i$$

Note: For the sake of simplicity, we still assume that the considered materials only reflect!

Rewriting the equation in operator form, the solution of the rendering equation is given by the Neumann series

$$L = (I - T)^{-1} L_e = \sum_{i=0}^{\infty} T^i L_e$$

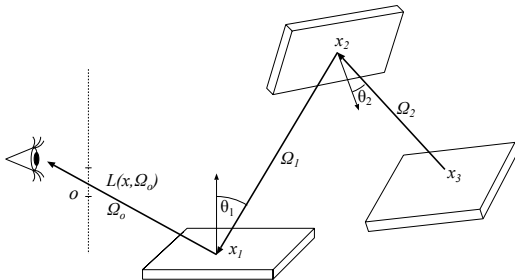
Since in general there is no analytical solution for the integral in the operator T , we have to do numerical integration for a full solution.

Problem: “Standard methods” are not well suited

- ▶ Integrand has discontinuities and is unevenly distributed $\rightarrow$ BRDF and incoming light
- ▶ Costly 2D nested integral $\rightarrow$ solving up to depth l leads to $2l$ -dimensional problem

Example:

$$\begin{aligned}
 T^2 L_e &= \int_{\mathcal{H}_1} f_{BSDF}(\Omega_1, \mathbf{x}_1, \Omega_o) L_o(\mathbf{h}(\mathbf{x}, \Omega_1), -\Omega_1) \cos(\theta_1) d\omega_1 \\
 &= \int_{\mathcal{H}_1} f_{BSDF}(\Omega_1, \mathbf{x}_1, \Omega_o) \cos(\theta_1) \left(\int_{\mathcal{H}_2} f_{BSDF}(\Omega_2, \mathbf{x}_2, \Omega_1) L_e(\mathbf{h}(\mathbf{x}, \Omega_2), -\Omega_2) \cos(\theta_2) d\omega_2 \right) d\omega_1
 \end{aligned}$$





Numerical Integration



Before we try to solve the Rendering Equation, we start solving a simple 1D integral

$$I = \int_{\mathcal{V}} f(\mathbf{z}) d\mathbf{z}$$

of a function

$$f: \mathcal{V} \subset \mathbb{R}^d \rightarrow \mathbb{R}.$$

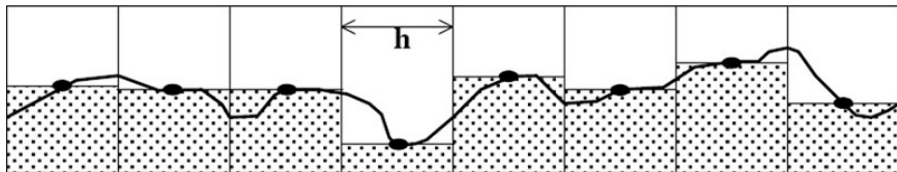
Numerical quadrature formulae take finite samples from the domain, evaluate the integrand for these samples and generate the integral as a weighted sum of the integrand values. The general form for the solution is

$$I \approx \hat{I} = \sum_{i=1}^n w(\mathbf{z}_i) f(\mathbf{z}_i),$$

where $\mathbf{z}_i$ are sample points from the domain $\mathcal{V}$ and $w(\mathbf{z}_i)$ are weights according to the chosen quadrature rule.

Example: Midpoint-Rule

$$\mathcal{V} = [a, b] \subset \mathbb{R}$$

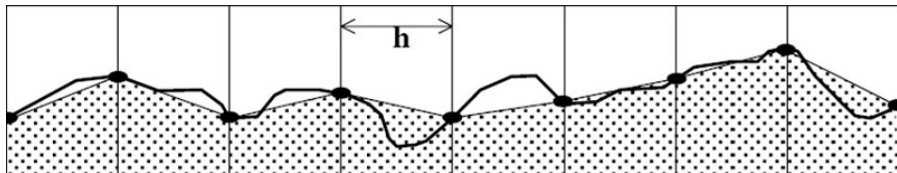


$$\begin{aligned}\hat{I} &= h \sum_{i=1}^n f\left(a + \left(i - \frac{1}{2}\right) h\right) \\ &= h \left[f\left(a + \frac{h}{2}\right) + f\left(a + \frac{3h}{2}\right) + \cdots + f\left(b - \frac{h}{2}\right) \right]\end{aligned}$$

Convergence : $\exists \xi \in [a, b]: \hat{I} - I = \frac{(b-a)^3}{24n^2} f''(\xi) = \mathcal{O}(n^{-2})$

Example: Trapezoidal-Rule

$$\mathcal{V} = [a, b] \subset \mathbb{R}$$



$$\begin{aligned}\hat{I} &= h \sum_{i=1}^n \frac{1}{2} [f(a + (i-1)h) + f(a + ih)] \\ &= h \left[\frac{1}{2} f(a) + f(a+h) + f(a+2h) + \cdots + f(b-h) + \frac{1}{2} f(b) \right]\end{aligned}$$

Convergence : $\exists \xi \in [a, b]: \hat{I} - I = \frac{(b-a)^3}{12n^2} f''(\xi) = \mathcal{O}(n^{-2})$

So depending on the used quadrature rule, we get a convergence rate of

$$\hat{I} - I = \sum_{i=1}^n w(z_i) f(z_i) - \int_{\mathcal{V}} f(z) dz = \mathcal{O}(n^{-k}),$$

e.g. for the midpoint rule $k = 2$. This was the 1D case where $\mathcal{V} \subset \mathbb{R}$.

Problem: For the d -dimensional case, i.e. $\mathcal{V} \subset \mathbb{R}^d$, the error becomes

$$\hat{I} - I = \sum_{i=1}^n w(z_i) f(z_i) - \int_{\mathcal{V}} f(z) dz = \mathcal{O}(n^{-k/d}),$$

since we have to sample each dimension equally to obtain the same error. This is called the “**Curse of Dimensionality**”.

One consequence of this “curse” is the way how the number of samples increases while reducing the error.

Lemma (Error Reduction)

Let the error of a quadrature rule using n samples be $E = |\hat{I} - I| = \mathcal{O}(n^{-\alpha})$.

Reducing the error by a factor of $\delta < 1$, at least $\left(\frac{1}{\delta}\right)^{\frac{1}{\alpha}} \cdot n$ samples have to be drawn:

$$E \rightarrow \delta \cdot E \quad \Rightarrow \quad n \rightarrow \left(\frac{1}{\delta}\right)^{\frac{1}{\alpha}} \cdot n$$

Proof.

Consider the error when using more samples:

$$\mathcal{O}((m \cdot n)^{-\alpha}) = c \cdot (m \cdot n)^{-\alpha} = c \cdot m^{-\alpha} \cdot n^{-\alpha} = m^{-\alpha} \cdot \mathcal{O}(n^{-\alpha})$$

By limiting this factor by δ , we get the desired result:

$$m^{-\alpha} \leq \delta \quad \Leftrightarrow \quad m^{\alpha} \geq \frac{1}{\delta} \quad \Leftrightarrow \quad m \geq \left(\frac{1}{\delta}\right)^{\frac{1}{\alpha}}$$



Problem: The Rendering Equation is recursively coupled. So in theory, a perfect solution would be “infinite”-dimensional.

But since the integral operator is a contraction, only computing up to some depth l leads to good approximations with an error of

$$E = \|\hat{L}_o - L_o\| = \mathcal{O}(n^{-k/2l}).$$

For very good solutions, l would be very large. And since k is constant here but not l , the total convergence behavior of the method is exponential due to the previous lemma:

$$E \rightarrow \frac{1}{2} \cdot E \quad \Rightarrow \quad n \rightarrow 2^{2l/k} \cdot n.$$

So for halving the error, one needs exponentially more samples. This is not feasible anymore!



Before we go further, we take a closer look at random variables. Consider a random variable

$$Z$$

over a domain $\mathcal{V} \subset \mathbb{R}$. Its Probability Density Function (PDF)

$$p_Z(z)$$

describes the likelihood that the random variable Z will have the value z (we define $z \notin \mathcal{V} \Rightarrow p_Z(z) := 0$). For example, the likelihood that Z falls into the interval $[a, b]$ is

$$\Pr(a \leq Z \leq b) = \int_a^b p_Z(z') dz'.$$

The Cumulative Distribution Function (CDF) is then given by

$$P_Z(z) = \Pr(Z \leq z) = \int_{-\infty}^z p_Z(z') dz'.$$



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Using this definition, the PDF can be seen as the derivative of the CDF:

$$p_Z(z) = \frac{d}{dz} P_Z(z).$$

Properties:

- ▶ PDF and CDF are both non-negative since they are related to probabilities
- ▶ PDF integrates to 1

$$\int_{-\infty}^{\infty} p_Z(z') dz' = 1$$

- ▶ CDF lies in the interval $[0, 1]$

$$\forall z: 0 \leq P_Z(z) \leq 1$$



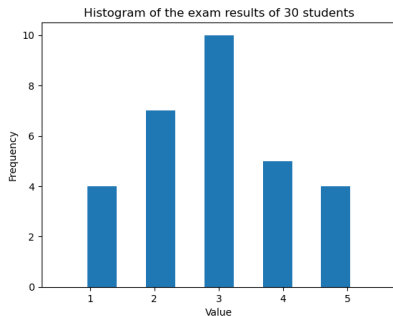
The **expected value** of a discrete random variable X with the set of possible values A and probability mass function $p(x)$ is defined by

$$\mathbb{E}(X) = \sum_{x \in A} x \cdot p(x). \quad (1)$$

The expected value of a random variable X is also called the mean, or the mathematical expectation, or simply the expectation of X .

In the example this results in $\mathbb{E}(X) = 1 \cdot \frac{4}{30} + 2 \cdot \frac{7}{30} + 3 \cdot \frac{10}{30} + 4 \cdot \frac{5}{30} + 5 \cdot \frac{4}{30} = \frac{88}{30}$

Note, that if each value x of X is weighted by $p(x) = P(X = x)$, then $\sum_{x \in A} x \cdot p(x)$ is nothing but the weighted average of X . Similarly, if we think of a unit mass distributed along the real line at the points of A so that the mass at $x \in A$ is $P(X = x)$, then $\mathbb{E}(X)$ is the center of gravity.





If X is a continuous random variable with probability density function p , the **expected value** of X is defined by

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x \cdot p(x) d\mu(x). \quad (2)$$

The expected value of X is also called the mean or simply the expectation of X , and as in the discrete case, sometimes it is denoted by $\mathbb{E}(X)$ or $\mathbb{E}[X]$.

Strong law of large numbers The intuition behind the expected value of random variable is to capture the mean of an infinitely large number of realizations of X . This is formalized by

$$\text{Prob} \left(\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N x_i = \mathbb{E}(X) \right) = 1, \quad (3)$$

where x_i are independent and identically distributed random variables following the same distribution as X .



Given an integral over an arbitrary domain D with measure μ

$$F = \int_D f(x) d\mu(x), \quad (4)$$

the idea of Monte Carlo integration is to express the integral as an expectation over some probability density $q(x)$

$$F = \int_D \frac{f(x)}{q(x)} q(x) d\mu(x) = \mathbb{E} \left[\frac{f(X)}{q(X)} \right] \approx \frac{1}{N} \sum_{i=1}^N \frac{f(x_i)}{q(x_i)}, \quad (5)$$

which is valid if $f(x) \neq 0 \Rightarrow q(x) > 0, \forall x \in D$.

Definition (Monte-Carlo Estimator¹)

The *Monte-Carlo Estimator* for solving the integral problem $F = \int_D f(x) d\mu(x)$ is defined as

$$\langle F \rangle_N = \frac{1}{n} \sum_{i=1}^n \frac{f(x_i)}{q(x_i)}$$

where $q(x)$ is a probability density in D and the samples x_i are selected according to this probability density.

¹I. M. Sobol. *Die Monte-Carlo-Methode*. Deutscher Verlag der Wissenschaften, 1991.

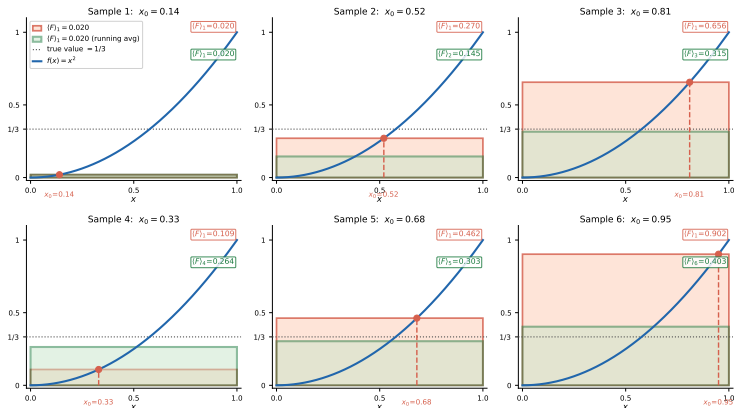


Monte-Carlo Quadrature – Visualising the Primary Estimator



Each single-sample MC estimate $\langle F \rangle_1 = \frac{f(x_0)}{q(x_0)}$ for $\int_0^1 x^2 dx$ with $q \equiv 1$ equals the **area of the rectangle** of width 1 and height $f(x_0) = x_0^2$:

Primary MC Estimator $\langle F \rangle_1 = x_0^2$ (red) and Running Average $\langle F \rangle_N$ (green) for $\int_0^1 x^2 dx = 1/3$



The dotted line marks the true value $\frac{1}{3}$. Averaging N such rectangle areas yields the MC estimate $\langle F \rangle_N$.

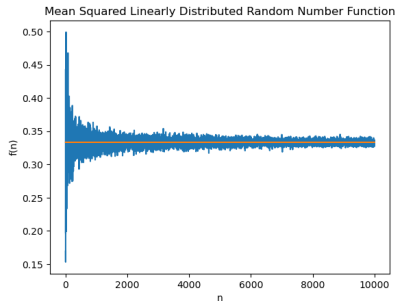
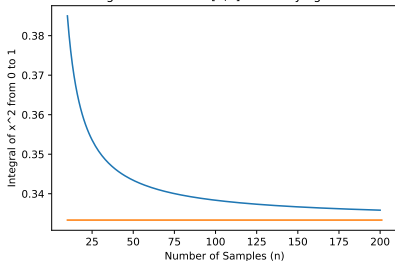


Definition (Monte Carlo estimate and Primal Monte Carlo estimator)

Any particular realization of the random variable $\langle F \rangle_N$ is called a **Monte Carlo estimate**. $\langle F \rangle_1 = \frac{f(X)}{q(X)}$ is called **primal Monte Carlo estimator**.

Example: Comparison between Standard Numerical Quadrature and Monte Carlo for $\int_0^1 x^2 dx \approx \frac{1}{n} \sum_{i=1}^n \frac{x_i^2}{q(x_i)}$ with $q(x_i) = 1$

Riemannian Integral of x^2 over $[0,1]$ with varying number of samples





A measure for the deviation between an estimator and its expectation value is its **variance**.

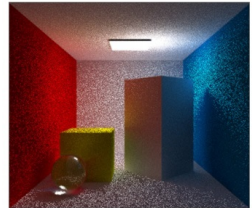
Theorem

*The **Variance of any Monte Carlo Estimator** $\langle F \rangle_N$ is proportional to $1/N$, regardless of the choice of $f(x)$ or $q(x)$.*

Proof.

$$\begin{aligned}\mathbb{V}[\langle F \rangle_N] &= \mathbb{V}\left[\frac{1}{N} \sum_{i=1}^N \frac{f(X_i)}{q(X_i)}\right] = \frac{1}{N^2} \mathbb{V}\left[\sum_{i=1}^N \frac{f(X_i)}{q(X_i)}\right] \\ &= \frac{1}{N^2} \sum_{i=1}^N \mathbb{V}\left[\frac{f(X_i)}{q(X_i)}\right] = \frac{1}{N} \mathbb{V}[\langle F \rangle_1]\end{aligned}$$

□



large variance



small variance

Theorem (Central Limit Theorem (CLT))

Let $X_1, \dots, X_N$ be independent and identically distributed (i.i.d.) random variables with mean $\mu = \mathbb{E}[X_i]$ and finite variance $\sigma^2 = \mathbb{V}[X_i] < \infty$. Then, as $N \rightarrow \infty$:

$$\frac{\bar{X}_N - \mu}{\sigma/\sqrt{N}} \xrightarrow{d} \mathcal{N}(0, 1), \quad \bar{X}_N := \frac{1}{N} \sum_{i=1}^N X_i.$$

That is, $\bar{X}_N$ is asymptotically normally distributed: $\bar{X}_N \sim \mathcal{N}(\mu, \sigma^2/N)$.

Recall: For a normal distribution, the probability of the estimate lying within k standard deviations of the true mean is:

Interval	Probability	Common name
$ \bar{X}_N - \mu < \sigma/\sqrt{N}$	≈ 0.683	1σ -rule
$ \bar{X}_N - \mu < 2\sigma/\sqrt{N}$	≈ 0.954	2σ -rule
$ \bar{X}_N - \mu < 3\sigma/\sqrt{N}$	≈ 0.997	3σ -rule

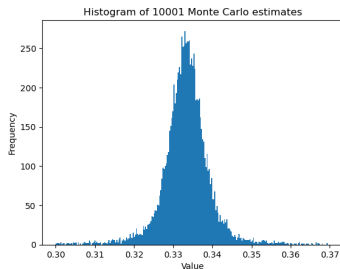
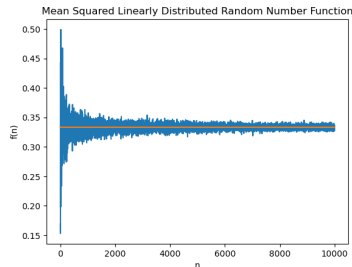
Key observation: The convergence rate $\sigma/\sqrt{N}$ is **independent of the dimension** of the problem — a decisive advantage we will exploit on the next slide.



Specialising to **uniform sampling** ($q(x) = 1/|D|$) on a compact domain $D \subset \mathbb{R}^d$ with Lebesgue measure $|D| := \int_D dx$, the estimator reduces to $\frac{|D|}{n} \sum_{i=1}^n f(x_i)$ and the **Central Limit Theorem** implies that the Monte Carlo Estimator $\langle F \rangle_N$ will asymptotically be normally distributed, with mean $F = \mathbb{E}[\langle F \rangle_1] = \int_D f(x) dx$ and standard deviation $\sigma = \frac{\sqrt{\mathbb{V}[\langle F \rangle_1]}}{\sqrt{n}}$. Therefore,

$$P(|\langle F \rangle_N - F| < 3\sigma) = 0.997$$

Thus with 0.997 confidence level, we can say that the (probabilistic) error bound of the Monte-Carlo quadrature is $|\int_D f(x) dx - \frac{|D|}{n} \sum_{i=1}^n f(x_i)| < 3\sigma = \frac{3\sqrt{\mathbb{V}[\langle F \rangle_1]}}{\sqrt{n}} = \mathcal{O}\left(n^{-\frac{1}{2}}\right)$.

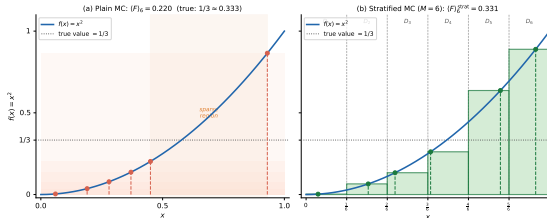


Core idea: On the full domain D , f may vary a lot $\Rightarrow$ high sample variance. On a *small* stratum D_m , f barely changes $\Rightarrow$ within-stratum variance $\text{Var}_m(f) \ll \text{Var}(f)$.

Definition (Stratified Sampling)

Partition D into M disjoint **strata** $D_1, \dots, D_M$ with $D = \bigcup_m D_m$. Draw n_m samples *independently and uniformly* from each D_m . The **stratified estimator** is

$$\langle F \rangle_N^{\text{strat}} = \sum_{m=1}^M \frac{|D_m|}{n_m} \sum_{j=1}^{n_m} f(x_{m,j}), \quad N = \sum_{m=1}^M n_m.$$



In panel (b), $f(x) = x^2$ barely changes within each narrow stratum D_m — the per-stratum

The variance of the stratified estimator decomposes per stratum:

$$\mathbb{V}[\langle F \rangle_N^{\text{strat}}] = \sum_{m=1}^M \frac{|D_m|^2}{n_m} \text{Var}_m(f),$$

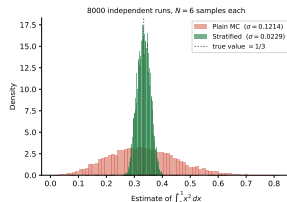
where $\text{Var}_m(f)$ is the variance of f within stratum D_m .

Stratified sampling replaces one high-variance global estimate with M low-variance local ones.

Theorem

For equal strata ($|D_m| = |D|/M$) and equal allocation ($n_m = N/M$):

$$\mathbb{V}[\langle F \rangle_N^{\text{strat}}] \leq \mathbb{V}[\langle F \rangle_N^{\text{plain}}].$$



Proof.

With equal strata and allocation: $\mathbb{V}[\langle F \rangle_N^{\text{strat}}] = \frac{|D|^2}{N} \cdot \frac{1}{M} \sum_m \text{Var}_m(f).$

The **law of total variance** gives

$$\mathbb{V}[f(X)] = \underbrace{\frac{1}{M} \sum_m \text{Var}_m(f)}_{\text{within-stratum}} + \underbrace{\frac{1}{M} \sum_m (\mu_m - \mu)^2}_{\geq 0} \geq \frac{1}{M} \sum_m \text{Var}_m(f),$$

hence $\mathbb{V}[\langle F \rangle_N^{\text{strat}}] \leq \frac{|D|^2}{N} \mathbb{V}[f(X)] = \mathbb{V}[\langle F \rangle_N^{\text{plain}}].$ □

Take-away: Choose strata so f is *approximately constant within* each stratum — this minimises $\text{Var}_m(f)$ and maximises the reduction.



Take away:

- ▶ Quadrature methods use fixed nodes and weights to approximate the integral. These methods can be very accurate for low-dimensional integrals but can become computationally expensive as the number of dimensions increases.
- ▶ Monte Carlo integration can be much more efficient than quadrature methods in high-dimensional cases, where the number of points required for quadrature methods to achieve a given level of accuracy grows exponentially with the number of dimensions.
- ▶ Monte Carlo integration provides a natural way to estimate the error in the integral estimate using statistical techniques, such as the standard deviation of the sample mean.
 - ▶ **Convergence of Variance independent on Dimension**
 - ▶ To **half** the variance of a given Monte Carlo estimator the number of random samples N needs to be **doubled**.
 - ▶ To **half** the standard deviation $\sqrt{\mathbb{V}[\langle F \rangle_N]}$ the number of random samples N must be **quadrupled**.
- ▶ **Flexibility:** Monte Carlo integration can be used to estimate integrals in any number of dimensions, making it suitable for problems that are difficult or impossible to solve using quadrature methods.



Definition (Consistency of an Estimator)

Let E_N be an N -sample estimator of some true value E , then E_N is **consistent** if and only if it approaches E as N approaches infinity:

$$\forall \delta > 0 : \lim_{N \rightarrow \infty} \text{Prob}(|E_N - E| > \delta) = 0.$$

In the case of the Monte Carlo estimators the weak law of large numbers guarantees their consistency.

Definition (Bias of an Estimator)

The **bias** of an estimator E_N is the deviation of its expectation $\mathbb{E}[E_N]$ from the true value E :

$$\text{Bias}[E_N] = \mathbb{E}[E_N] - E.$$

An estimator is called **unbiased** if its bias is zero, i.e., its expectation matches the true value.

Note, that in general an unbiased estimator is not necessarily consistent because unbiasedness does not guarantee that the estimator converges in probability to any value at all as N approaches infinity.



Definition (Efficiency)

Let $\hat{f}$ be a Monte-Carlo Estimator. The *efficiency* of $\hat{f}$ is defined as

$$\epsilon(\hat{f}) = \frac{1}{\mathbb{V}(\hat{f}) \cdot T(\hat{f})}$$

where $\mathbb{V}(\hat{f})$ is the variance and $T(\hat{f})$ the computation time of $\hat{f}$.

So an algorithm is better than another one if $\mathbb{V}[F]$

- ▶ it has lower variance at the same computation time
- ▶ it is faster at the same variance

→ We first continue with an important computation time reduction technique called **Russian Roulette** and then with **importance sampling**.



Consider a sequence $Y_1, Y_2, \dots$ with finite limit Y_∞ . How can we estimate the limit after having seen only a finite number of elements? $\rightarrow$ Use **Russian Roulette**²

Algorithm Russian Roulette Estimator

Require: Probability $\alpha \in [0, 1)$, initial estimate Y_0

- 1: Set $\hat{Y} = Y_0$
 - 2: **for** $t = 1, \dots$ **do**
 - 3: With probability α : halt and return $\hat{Y}$
 - 4: Compute Y_t and $\hat{Y} = \hat{Y} + (1 - \alpha)^{-t}(Y_t - Y_{t-1})$
 - 5: **end for**
-

Note, that at each subsequent iteration, the algorithm produces a Bernoulli trial with a probability of success α . If the trial was successful, then the algorithm stops and returns the current estimate. If trial was not successful, then the algorithm continues and updates the current estimate $\hat{Y}$ with a new element of the sequence³:

$$\hat{Y} = \hat{Y} + (1 - \alpha)^{-t}(Y_t - Y_{t-1})$$

²G. E. Forsythe and R. A. Leibler. "Matrix inversion by a Monte Carlo method". In: *Mathematics of Computation* 4.31 (1950), pp. 127–129.

³F. Pedregosa. *The Russian Roulette: An Unbiased Estimator of the Limit*. 2022. URL: <http://fa.bianp.net/blog/2022/russian-roulette/>.



Theorem (The Russian Roulette estimator is unbiased)

The Russian Roulette is an unbiased estimator of the limit of the sequence. That is, if E denotes the expectation with respect to the randomness of the halting time, then we have $\mathbb{E}[\hat{Y}] = Y_\infty$.

Proof.

Let T denote the total number of iterations performed by the estimator $\hat{Y}$ and $\mathbb{1}_{\{i < T\}}$ denote a variable that is 1 if $i < T$ and 0 otherwise. Using this notation, we can write the Russian roulette as the infinite sum

$$\hat{Y} = Y_0 + \sum_{i=1}^{\infty} \mathbb{1}_{\{i < T\}} (1 - \alpha)^{-i} (Y_i - Y_{i-1}),$$

as all terms after T will be zero.





cont.

Since we know that T follows a geometric distribution with parameter α , we can use this to compute $\mathbb{E}[\mathbb{1}_{\{i < T\}}] = P(i < T) = 1 - P(T \leq i)$. This last term is the cumulative distribution function of a geometric distribution, which is $1 - (1 - \alpha)^i$, and so $\mathbb{E}[\mathbb{1}_{\{i < T\}}] = (1 - \alpha)^i$. Finally, taking expectations in the above formula and using this last fact we have the desired unbiasedness:

$$\begin{aligned}\mathbb{E}[\hat{Y}] &= \mathbb{E}\left[Y_0 + \sum_{i=1}^{\infty} \mathbb{1}_{\{i < T\}} (1 - \alpha)^{-i} (Y_i - Y_{i-1})\right] \\ &= Y_0 + \sum_{i=1}^{\infty} \mathbb{E}[\mathbb{1}_{\{i < T\}}] (1 - \alpha)^{-i} (Y_i - Y_{i-1}) \\ &= Y_0 + \sum_{i=1}^{\infty} (Y_i - Y_{i-1}) = Y_{\infty}.\end{aligned}$$

□



The solution of the rendering equation in form of the Neumann series is given by

$$L = (I - T)^{-1} L_e = \sum_{i=0}^{\infty} T^i L_e$$

Let

$$Y_t := \sum_{i=0}^t T^i L_e \quad \text{we get} \quad L = \lim_{t \rightarrow \infty} Y_t.$$

We further observe that

$$Y_t - Y_{t-1} = \sum_{i=0}^t T^i L_e - \sum_{i=0}^{t-1} T^i L_e = T^t L_e.$$

Plugging this into the update for the current estimate in the Russian Roulette estimator we get

$$\hat{Y} = \hat{Y} + (1 - \alpha)^{-t} (Y_t - Y_{t-1}) = \hat{Y} + (1 - \alpha)^{-t} T^t L_e$$



Using this result

$$\hat{Y} = \hat{Y} + (1 - \alpha)^{-t}(Y_t - Y_{t-1}) = \hat{Y} + (1 - \alpha)^{-t}T^t L_e$$

in the Russian Roulette algorithm results in

Algorithm Russian Roulette von Neumann series estimator

Require: Probability $\alpha \in [0, 1)$

- 1: Set $\hat{Y} = L_e$
 - 2: **for** $t = 1, \dots$ **do**
 - 3: With probability α : halt and return $\hat{Y}$
 - 4: Update $\hat{Y} = \hat{Y} + (1 - \alpha)^{-t}T^t L_e$
 - 5: **end for**
-

Interpretation Instead of taking all terms of the von Neumann series into account, we'll use the termination probability α to decide whether to continue the sum or terminate it. To account for the neglected term, we multiply the contribution of the remaining path lengths by an appropriate weight, in order to maintain the overall balance.



Example: Stopping after 3 steps

$$\begin{aligned}L &= \sum_{i=0}^{\infty} T^i L_e \\&= L_e + \frac{1}{1-\alpha} \left(\sum_{i=1}^{\infty} T^i L_e \right) \\&= L_e + \frac{1}{1-\alpha} \left(T L_e + \frac{1}{1-\alpha} \left(\sum_{i=2}^{\infty} T^i L_e \right) \right) \\&= L_e + \frac{1}{1-\alpha} \left(T L_e + \frac{1}{1-\alpha} \left(T^2 L_e + \frac{1}{1-\alpha} \left(\sum_{i=3}^{\infty} T^i L_e \right) \right) \right) \\&= L_e + \frac{1}{1-\alpha} \left(T L_e + \frac{1}{1-\alpha} \left(T^2 L_e + \frac{1}{1-\alpha} (T^3 L_e) \right) \right) \\&= L_e + \frac{1}{1-\alpha} T L_e + \frac{1}{(1-\alpha)^2} T^2 L_e + \frac{1}{(1-\alpha)^3} T^3 L_e\end{aligned}$$



It should be noted that in the context of computer graphics, it is often only the individual steps that are looked at and analyzed:

A random variable is used to stop a path of length i with probability α and to continue it with $(1 - \alpha)$. This is realized by drawing a random number ξ_i using a uniform distribution on the interval $[0, 1]$ and defining a novel random variable for the contribution of longer path length by setting

$$F' = \begin{cases} \frac{F}{(1-\alpha)}, & \text{if } \alpha < \xi_i, \\ 0, & \text{otherwise.} \end{cases}$$

This delivers an **unbiased estimator**, since

$$\mathbb{E}(F') = \frac{\mathbb{E}(F)}{(1-\alpha)}(1-\alpha) + 0 \cdot \alpha = \mathbb{E}(F).$$

Unfortunately, in each step the **variance increases**:

$$\begin{aligned} \mathbb{V}(F') &= \mathbb{E}((F')^2) - \mathbb{E}(F')^2 = \frac{1}{(1-\alpha)}\mathbb{E}(F^2) - \mathbb{E}(F)^2 \\ &= \frac{1}{(1-\alpha)}\mathbb{E}(F^2) - \mathbb{E}(F^2) + \mathbb{E}(F^2) - \mathbb{E}(F)^2 \\ &= \frac{1}{(1-\alpha)}\mathbb{E}(F^2)(1 - (1-\alpha)) - \mathbb{V}(F) = \frac{\alpha}{(1-\alpha)}\mathbb{E}(F^2) + \mathbb{V}(F) \end{aligned}$$



An analysis of the overall variance is given by the following theorem (Pedregosa, 2022:)

Theorem

The variance of the Russian Roulette estimate is

$$\mathbb{E}[\|\hat{Y} - Y_\infty\|_F^2] = \sum_{i=0}^{\infty} \underbrace{\alpha(1-\alpha)^{-(i+1)}}_{\text{diverging}} \underbrace{\|\hat{Y}_i - Y_\infty\|_F^2}_{\text{converging}}.$$

Note, that although we've assumed Y_t converges to Y_∞ as $t \rightarrow \infty$ and so these terms asymptotically vanish, this is not enough for the variance to be finite. For the variance to be finite, it's necessary that the terms $\|Y_\infty - Y_i\|_F^2$ need to converge at a speed faster than $O((1-\alpha)^t)$.

Summary (Russian Roulette)

In general the Russian Roulette will often result in useless estimators with infinite variance. Unless one is extremely careful and chooses fast-converging sequences like in most computer graphics scenarios or large average number of iterations (small α), then the variance of this estimator will be infinite.



Question: How to choose the threshold α ?

Idea: As we will see in the context of path tracing, the weight

$$w(\hat{\Omega}_i) = \frac{f_{BRDF}(\hat{\Omega}_i, \mathbf{x}, \Omega_o) \cos(\hat{\theta}_i)}{p(\mathbf{x}, \hat{\Omega}_i)} \in \mathbb{R}^d$$

has a great impact on the sum of samples. Since it is the only value which we know for sure, we set

$$\alpha = \max_{1 \leq i \leq d} w_i(\hat{\Omega}_i).$$

Therefore, the threshold α accounts for the strongest component of the light that gets reflected to the viewer. If this value is small, the reflected portion of light is also very small (or even smaller)!



Importance sampling is a well-known method of Monte-Carlo integration to **reduce variance**. The basic idea is to use a non-uniform distribution q that reduces the variance of the **Monte Carlo estimator** $\langle F \rangle_N = \frac{1}{N} \sum_{i=1}^N \frac{f(X_i)}{q(X_i)}$.

Question: But how should we choose q to minimize the variance ?

Lemma (Importance Sampling)

Let $F = \int_D f(x) d\mu(x)$. If we choose $q(x) = \frac{f(x)}{F}$ than the variance of $\langle F \rangle_1$ is zero.

Proof.

$$\langle F \rangle_1 = \frac{f(X)}{q(X)} = \frac{f(X)}{f(X)/F} = F$$

□

Note: This implies that we already have to know the solution for the integral in order to solve it using Monte-Carlo quadrature! Unfortunately, to choose q in such a way we have to know the integral we want to solve in advance.

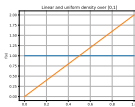


Importance Sampling



Problem: As noted before, this implies that we already know the integral!

Idea: Choose p “as similar as possible” to f in order to minimize the variance of the primal estimator $\langle F \rangle_1 = \frac{f(X)}{q(X)}$.



Example 1 Integrating $f(x) = x^2$ over $[0, 1]$:

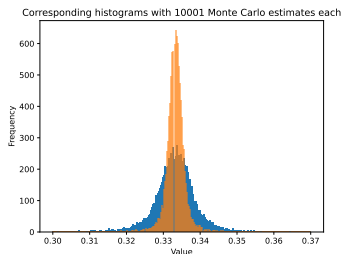
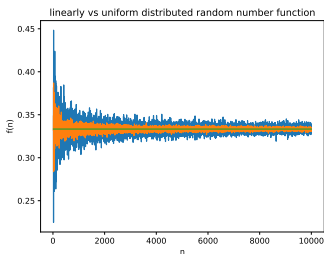


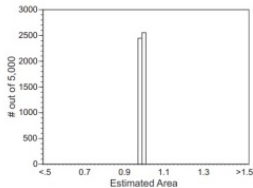
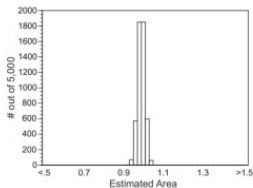
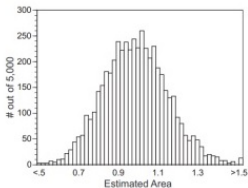
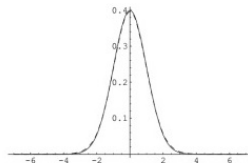
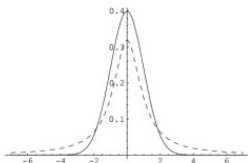
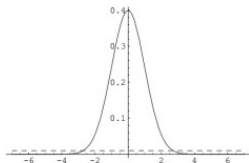
Figure: Comparison between Monte Carlo approximation of $F = \int_0^1 x^2 dx$ using uniform distribution (blue) and linear distribution $q(x) = 2x$ (orange).



Importance Sampling



Example 2: Area estimation of a normal distribution using Importance Sampling.





Example 3: Integrating $f(x) = x^2$ over $[0, 1]$ with inappropriate density.

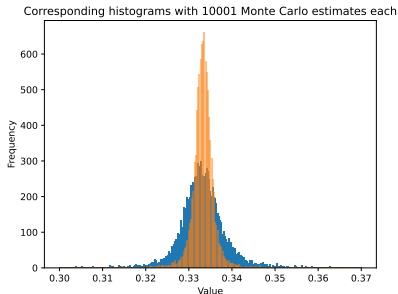
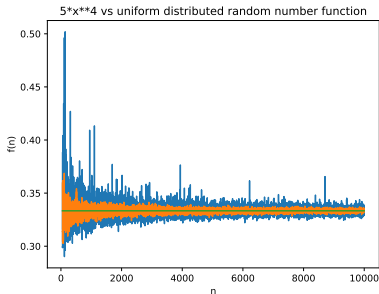
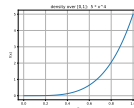


Figure: Comparison between Monte Carlo approximation of $F = \int_0^1 x^2 dx$ using uniform distribution (orange) and polynomial distribution $q(x) = 5x^4$ (blue).



Multiple Importance Sampling (MIS) is a technique used to reduce the variance of Monte Carlo integration estimates when there are multiple sampling strategies available.

Definition

The MIS estimate is defined as:

$$\langle F \rangle = \sum_{i=1}^m \frac{1}{n_i} \sum_{j=1}^{n_i} \frac{w_i(x_{i,j}) f(x_{i,j})}{p_i(x_{i,j})}$$

where the weights $w_i(x_{ij})$ are such that

$$f(x) \neq 0 \Rightarrow \sum_{i=1}^m w_i(x) = 1 \quad \text{and} \quad p_i(x) = 0 \Rightarrow w_i(x) = 0.$$

In this combination scheme, sampling method i uses probability density function $p_i(x)$ to generate n_i number of random samples $\{x_{i,j}\}$, ($j = 1, \dots, n_i$). If we have m sampling techniques, the total number of samples is $\sum_{i=1}^m n_i = N$. When the sample numbers n_i are fixed deterministically, the approach is called **multi-sample**.



Lemma

1. *The MIS estimator F is unbiased.*
2. *Assuming that the sampling strategies can be assumed to be independent random variables the variance of the MIS estimator is*

$$\mathbb{V}[F] = \sum_{i=1}^m \int \frac{w_i^2(x) f^2(x)}{n_i p_i(x)} dx - \sum_{i=1}^m \frac{\mu_i^2}{n_i}$$

Proof.

1. We show that its expected value μ is equal to integral F :

$$\begin{aligned} \mu &= \mathbb{E}[F] = \sum_{i=1}^m \frac{1}{n_i} \sum_{j=1}^{n_i} \int_D \frac{w_i(x) f(x)}{p_i(x)} p_i(x) dx \\ &= \int_D \sum_{i=1}^m w_i(x) f(x) dx = \int_D f(x) dx. \end{aligned}$$



cont.

2. Let

$$F_{i,j} = \frac{w_j(x_{i,j})f(x_{i,j})}{p_i(x_{i,j})} \text{ and } \mu_i = \mathbb{E}[F_{i,j}] = \int_D w_i(x)f(x)dx$$

be its expected value. Taking into account that the $x_{i,j}$ are sampled independently the variance of the Multiple Importance Sampling (MIS) estimator is given by:

$$\begin{aligned}\mathbb{V}[F] &= V\left[\sum_{i=1}^m \frac{1}{n_i} \sum_{j=1}^{n_i} F_{i,j}\right] \quad \underbrace{\text{no covariance terms!}} \quad \sum_{i=1}^m \frac{1}{n_i^2} \sum_{j=1}^{n_i} \mathbb{V}[F_{i,j}] \\ &= \left(\sum_{i=1}^m \frac{1}{n_i^2} \sum_{j=1}^{n_i} \mathbb{E}[F_{i,j}^2]\right) - \left(\sum_{i=1}^m \frac{1}{n_i^2} \sum_{j=1}^{n_i} \mathbb{E}[F_{i,j}]^2\right) \\ &= \left(\sum_{i=1}^m \frac{1}{n_i^2} \sum_{j=1}^{n_i} \int_D \frac{w_i^2(x)f^2(x)}{p_i^2(x)} p_i(x) dx\right) - \left(\sum_{i=1}^m \frac{1}{n_i^2} n_i \mu_i^2\right) \\ &= \left(\int_D \sum_{i=1}^m \frac{w_i^2(x)f^2(x)}{n_i p_i(x)} dx\right) - \left(\sum_{i=1}^m \frac{\mu_i^2}{n_i}\right).\end{aligned}$$



How to choose the weights?

1. First we observe that the variance of the estimator depends on the combination weights $w_i(x)$.
2. The **balance heuristic**⁴ states that the weight of method i at domain point x should be proportional to the density of samples generated by method i in this point:

$$w_i(x) = \frac{\alpha_i p_i(x)}{p(\alpha, x)},$$

where $\alpha_i = \frac{n_i}{N}$ is the fraction of the samples allocated to method i , and $p(\alpha, x)$ is the mixture pdf:

$$p(\alpha, x) = \sum_{k=1}^m \alpha_k p_k(x).$$

Substituting this weighting function into the MIS estimator formulas, we obtain the **multisample balance heuristics estimator**:

$$F = \frac{1}{N} \sum_{i=1}^m \sum_{j=1}^{n_i} \frac{f(x_{i,j})}{p(\alpha, x_{i,j})}.$$

⁴E. Veach. *Robust Monte Carlo methods for light transport simulation*. Stanford University, 1998;
E. Veach and L. J. Guibas. "Optimally combining sampling techniques for Monte Carlo rendering". In:
Proceedings of the 22nd annual conference on Computer graphics and interactive techniques. ACM. 1995.



Lemma

The variance of the one-sample balance heuristics estimator is

$$\mathbb{V}[F] = \frac{1}{N} \left(\int_D \frac{f^2(x)}{p(\alpha, x)} dx - \sum_{i=1}^m \alpha_i \mu_i'^2 \right),$$

where $\mu_i' = \frac{\mu_i}{\alpha_i}$.

Proof.

Plugging the definition of the weights into the variance of the MIS estimator in the Lemma above delivers the result. □



Multiple Importance Sampling: the balance heuristic



Example 3: Integrating $f(x) = x^2$ over $[0, 1]$ with multiple importance sampling using a balance heuristic for two sampling strategies, one using uniform distribution (orange) and one using a polynomial distribution $q(x) = 5x^4$ (blue).

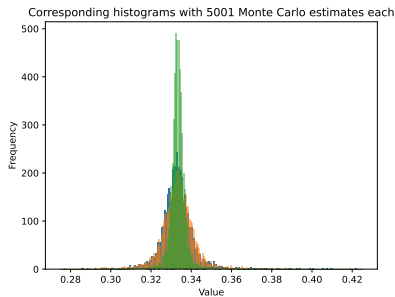
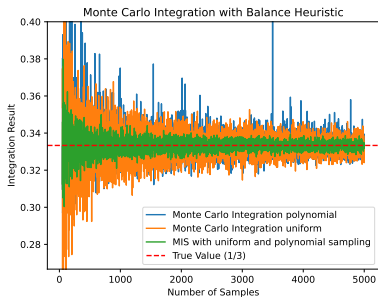
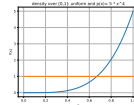


Figure: Comparison between Monte Carlo approximation of $F = \int_0^1 x^2 dx$ using uniform distribution (orange), polynomial distribution $q(x) = 5x^4$ (blue) and MIS combining the two using the balance heuristic.



Example 4: Rendering equation

$$L_o(\mathbf{x}, \Omega_o) = L_e(\mathbf{x}, \Omega_o) + \int_{\mathcal{H}_i} f_{BRDF}(\Omega_i, \mathbf{x}, \Omega_o) L_i(\mathbf{x}, \Omega_i) \cos(\theta_i) d\omega_i$$

Here, we basically want to solve an integral of the form

$$I = \int_{\mathcal{V}} f(\mathbf{z}) g(\mathbf{z}) d\mathbf{z},$$

If we sample according to both functions f and g the *Multi-Sample Monte-Carlo Estimator* is given as

$$F = \frac{1}{n_f} \sum_{i=1}^{n_f} w_f(\mathbf{z}_{f,i}) \frac{f(\mathbf{z}_{f,i}) g(\mathbf{z}_{f,i})}{p_f(\mathbf{z}_{f,i})} + \frac{1}{n_g} \sum_{i=1}^{n_g} w_g(\mathbf{z}_{g,i}) \frac{f(\mathbf{z}_{g,i}) g(\mathbf{z}_{g,i})}{p_g(\mathbf{z}_{g,i})}$$

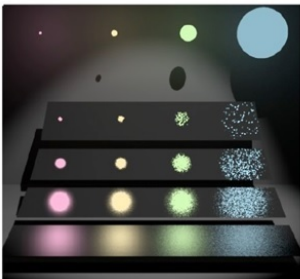
where $\mathbf{z}_{f,i}, \mathbf{z}_{g,i}$ are samples drawn from the PDFs p_f, p_g and w_f, w_g are weighting functions with $w_f(\mathbf{z}) + w_g(\mathbf{z}) = 1$ for all $\mathbf{z}$ where $f(\mathbf{z}) \neq 0 \neq g(\mathbf{z})$.



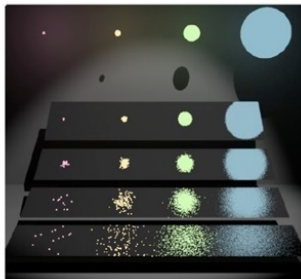
Example 4: Rendering equation

Sampling either the BRDF or the light sources leads to poor results in some scenarios:

Sample Light



Sample BRDF

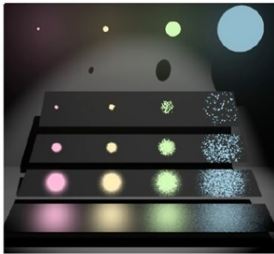


The four light sources of different size (with equal total energy) on the top should be visible on each plate. The four plates show glossy BRDFs of varying roughness.



Multiple Importance Sampling: Sampling from both, the light sources and BRDF simultaneously:

Sample Light



Sample BRDF



Result: better than either of the two strategies alone



This approach is called **Multiple Importance Sampling** and produces good results in all scenarios.



One obvious choice for the weights is the *Balance Heuristic*.
These weights can further be generalized in the *Power Heuristic*.

Definition (Power Heuristic)

$$w_f(\mathbf{z}) = \frac{(n_f p_f(\mathbf{z}))^\beta}{(n_f p_f(\mathbf{z}))^\beta + (n_g p_g(\mathbf{z}))^\beta} \quad , \quad w_g(\mathbf{z}) = \frac{(n_g p_g(\mathbf{z}))^\beta}{(n_f p_f(\mathbf{z}))^\beta + (n_g p_g(\mathbf{z}))^\beta}$$

Obviously, these weights fulfill the desired property:

$$\begin{aligned} w_f(\mathbf{z}) + w_g(\mathbf{z}) &= \frac{(n_f p_f(\mathbf{z}))^\beta}{(n_f p_f(\mathbf{z}))^\beta + (n_g p_g(\mathbf{z}))^\beta} + \frac{(n_g p_g(\mathbf{z}))^\beta}{(n_f p_f(\mathbf{z}))^\beta + (n_g p_g(\mathbf{z}))^\beta} \\ &= \frac{(n_f p_f(\mathbf{z}))^\beta + (n_g p_g(\mathbf{z}))^\beta}{(n_f p_f(\mathbf{z}))^\beta + (n_g p_g(\mathbf{z}))^\beta} \\ &= 1 \end{aligned}$$

This heuristic is called the **Power Heuristic**. Note, that $\beta = 1$ results in the Balance Heuristic.



Multiple Importance Sampling: the one-sample balance heuristic

- ▶ Note, that in the balance heuristic we need to fix the number of samples drawn by each strategy before sampling.
- ▶ In the **one-sample** version of MIS, the numbers of samples used by particular techniques are not determined before starting the sampling process, but for each sample, the sampling method itself is selected randomly with the probability of fraction parameter α_i .
- ▶ As this approach introduces additional randomization, its variance is larger than that of the multisample version, but can also be used in cases when the number of total samples is less than the number of combined methods.
- ▶ Let $p(\alpha, x) = \sum_{k=1}^m \alpha_k p_k(x)$ be a mixture pdf of the different sampling strategies then the **one-sample balance heuristic estimator**⁵ is given by

$$F = \frac{f(X)}{p(\alpha, X)}$$

with variance

$$\mathbb{V}[F] = \int \frac{f^2(x)}{p(\alpha, x)} dx - \mu^2$$

⁵E. Veach. *Robust Monte Carlo methods for light transport simulation*. Stanford University, 1998.



Sampling Strategies





Now, we go back to the problem of Importance Sampling.

Question: Given a 1D function g defined over $\mathcal{V}$ which is similar to the integrand f . How can we create samples with a probability density that is proportional to g ?

The probability density p_Z should be proportional to g such that it integrates to 1:

$$p_Z(z) = \frac{g(z)}{\int_{\mathcal{V}} g(z') dz'}.$$

Defined in this way, it integrates to the desired value:

$$\int_{\mathcal{V}} p_Z(z') dz' = \int_{\mathcal{V}} \frac{g(z')}{\int_{\mathcal{V}} g(z') dz'} dz' = \frac{\int_{\mathcal{V}} g(z') dz'}{\int_{\mathcal{V}} g(z') dz'} = 1.$$

This makes obvious why we should use non-optimal densities. If g was equal to f , then the construction of the probability density would require the integral of f .

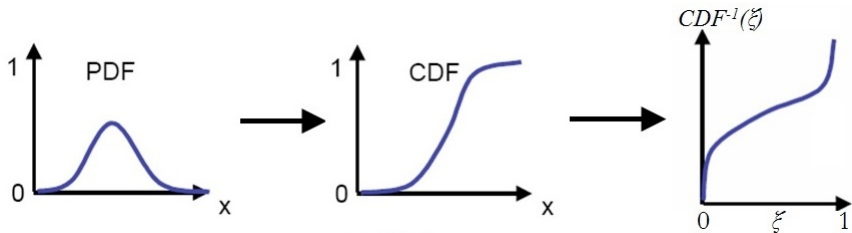
Note: A generalization to d dimensions can be made directly!



Task: Generate samples according to the Probability Density Function p_Z .

Algorithm Inversion Method

- 1: Compute the Cumulative Distribution Function (CDF) P_Z
 - 2: Invert the CDF P_Z
 - 3: Generate a sample ξ for the interval $[0, 1]$ uniformly, then its transformed version $P_Z^{-1}(\xi)$ is sampled according to the PDF p_Z
-





Theorem (Inversion Method)

Let ξ be a random variable distributed uniformly over $[0, 1]$. Let p_Z be a PDF defined over $\mathcal{V}$. Then, the random variable Z distributed over $\mathcal{V}$ with PDF p_Z has the identity

$$Z = P_Z^{-1}(\xi).$$

Proof.

We have to show that the PDFs of both random variables are equal:

$$\forall z \in \mathcal{V}: p_Z(z) = p_{P_Z^{-1}(\xi)}(z).$$

Equally, we can show this for the CDFs:

$$\begin{aligned} P_{P_Z^{-1}(\xi)}(z) &= \Pr(P_Z^{-1}(\xi) \leq z) \\ &= \Pr(\xi \leq P_Z(z)) && \text{since } P_Z \text{ is monotonic} \\ &= P_Z(z) && \text{since } \xi \text{ is uniform on the unit interval} \end{aligned}$$

Since the CDFs are equal, the PDFs also have to be equal. □



Example: Given a function g over the domain $\mathcal{V} = \mathbb{R}^{\geq 0}$:

$$g(z) = c \cdot e^{-az}$$

The PDF can be computed using

$$p_Z(z) = \frac{g(z)}{\int_{\mathcal{V}} g(z') dz'} = \frac{c \cdot e^{-az}}{\int_0^{\infty} c \cdot e^{-az'} dz'} = \frac{c \cdot e^{-az}}{\frac{c}{a}} = a \cdot e^{-az}.$$

The CDF is

$$P_Z(z) = \int_0^z a \cdot e^{-az'} dz' = 1 - e^{-az}.$$

Inverting the CDF leads to

$$P_Z^{-1}(z) = -\frac{1}{a} \cdot \ln(1 - z).$$

So our random variable Z can be expressed in terms of the uniformly distributed random variable ξ :

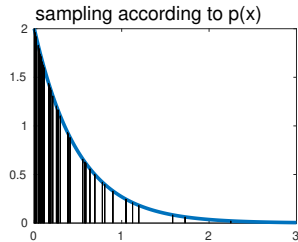
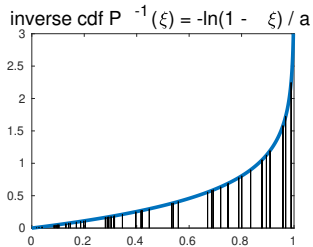
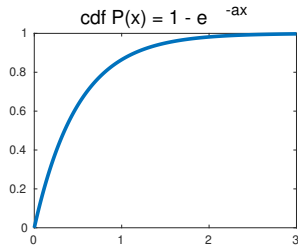
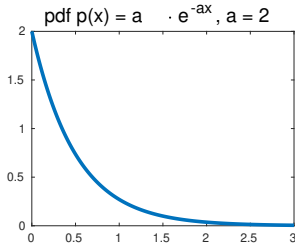
$$Z = P_Z^{-1}(\xi) = -\frac{1}{a} \cdot \ln(1 - \xi).$$



Inversion Method (Univariate)



Example:



Goal: Appropriate transformation of random distributions (i.e. PDFs)

→ generate “good” distributions from simple ones

Given:

- ▶ Random variable X with probability density function p_X
- ▶ Derived random variable $Y = t(X)$ where the transformation t is a bijective, monotonic continuous mapping

We want to find an expression for the PDF p_Y . First, we consider the CDFs:

$$\begin{aligned} P_Y(y) &= \Pr(Y \leq y) \\ &= \Pr(t(X) \leq t(x)) \\ &= \Pr(X \leq x) && \text{since } t \text{ is monotonic and bijective} \\ &= P_X(x) \end{aligned}$$

For the PDFs, the following holds (chain rule)

$$p_Y(y) = \frac{dP_Y(y)}{dy} = \frac{dP_X(x)}{dy} = \frac{dP_X(t^{-1}(y))}{dy} = \left| \frac{dt^{-1}(y)}{dy} \right| \cdot p_X(t^{-1}(y))$$

For the PDFs, the following holds (chain rule)

$$p_Y(y) = \frac{dP_Y(y)}{dy} = \frac{dP_X(x)}{dy} = \frac{dP_X(t^{-1}(y))}{dy} = \left| \frac{dt^{-1}(y)}{dy} \right| \cdot p_X(t^{-1}(y))$$

Theorem (Transformation of Variables (Univariate))⁶

Let X be a random variable over the domain $\mathcal{V}_X \subset \mathbb{R}$ with probability density function p_X . Furthermore, let Y be a derived random variable over the domain $\mathcal{V}_Y \subset \mathbb{R}$ with $Y = t(X)$ where t is a bijective, monotonic continuous mapping. Then the PDF of Y can be written as

$$p_Y(y) = \left| \frac{dt^{-1}(y)}{dy} \right| \cdot p_X(t^{-1}(y))$$

Example:

$$p_X(x) = 2x \quad , \quad 0 \leq x \leq 1 \quad ; \quad Y = 5X$$

$$p_Y(y) = \left| \frac{dt^{-1}(y)}{dy} \right| \cdot p_X(t^{-1}(y)) = \left| \frac{d}{dy} \left(\frac{1}{5}y \right) \right| \cdot 2 \left(\frac{1}{5}y \right) = \frac{1}{5} \cdot \frac{2}{5}y = \frac{2}{25}y$$

⁶A. Mood et al. *Introduction to the Theory of Statistics*. McGraw-Hill, 1974.

Problem: Typically, a desired p_Y (or P_Y) is given and we search for the appropriate values y . Consider the CDFs again:

$$P_Y(y) = \Pr(Y \leq y) = \Pr(t(X) \leq t(x)) = \Pr(X \leq x) = P_X(x)$$

From this, it follows that

$$y = P_Y^{-1}(P_X(x))$$

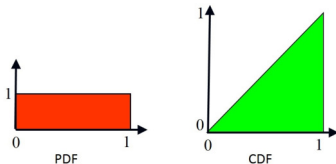
Theorem (Generalized Inversion Method)

Let X and Y be random variables over the domains $\mathcal{V}_X, \mathcal{V}_Y$ with probability density functions p_X, p_Y connected by the bijective, monotonic continuous mapping t . Then

$$Y = P_Y^{-1}(P_X(X))$$

Note: The standard Inversion Method is a special case of this one.

If X is uniformly distributed over $\mathcal{V}_X = [0, 1]$, then $P_X(x) = x$.



Example:

Domain	$\mathcal{V}_X = [0, \sqrt{2}]$	$\mathcal{V}_Y = [-\infty, 0]$
PDF	$p_X(x) = x$	$p_Y(y) = e^y$
CDF	$P_X(x) = \frac{1}{2}x^2$	$P_Y(y) = e^y$
Inverse CDF		$P_Y^{-1}(y) = \ln(y)$

The Generalized Inversion Method tells us

$$y = P_Y^{-1}(P_X(x)) = \ln\left(\frac{1}{2}x^2\right) = 2 \cdot \ln(x) - \ln(2)$$

Therefore, if the random variable X is distributed according to $p_X(x) = x$, the random variable $y = 2 \cdot \ln(X) - \ln(2)$ will have the PDF $p_Y(y) = e^y$.



If we would try to apply the Inversion Method described as before to the Rendering Equation, we would fail. The integral in the Rendering Equation has a 2-dimensional domain, but we only learned how to solve 1-dimensional integrals with the Inversion Method. So we need to generalize the concept:

Consider a d -dimensional random variable

$$\mathbf{Z}$$

over a domain $\mathcal{V} \subset \mathbb{R}^d$. Its Probability Density Function (PDF)

$$p_{\mathbf{Z}}(\mathbf{z}) = p_{\mathbf{Z}}(z_1, \dots, z_d) = \frac{\partial^d}{\partial z_1 \dots \partial z_d} P_{\mathbf{Z}}(\mathbf{z})$$

describes the likelihood that the random variable $\mathbf{Z}$ will have the value $\mathbf{z}$ (we define $\mathbf{z} \notin \mathcal{V} \Rightarrow p_{\mathbf{Z}}(\mathbf{z}) := 0$). The Cumulative Distribution Function (CDF) is given by

$$P_{\mathbf{Z}}(\mathbf{z}) = P_{\mathbf{Z}}(z_1, \dots, z_d) = \int_{-\infty}^{z_d} \dots \int_{-\infty}^{z_1} p_{\mathbf{Z}}(z'_1, \dots, z'_d) dz'_1 \dots dz'_d.$$

Since we made no assumption about the domain $\mathcal{V}$ in the Inversion Method (and the Generalized Inversion Method), it can be directly generalized. On the other hand the Transformation of Variables needs to be adjusted:

Theorem (Transformation of Variables (Multivariate))

Let $\mathbf{X}$ be a random variable over the domain $\mathcal{V}_{\mathbf{X}} \subset \mathbb{R}^d$ with probability density function $p_{\mathbf{X}}$. Furthermore, let $\mathbf{Y}$ be a derived random variable over the domain $\mathcal{V}_{\mathbf{Y}} \subset \mathbb{R}^d$ with $\mathbf{Y} = \mathbf{t}(\mathbf{X})$ where $\mathbf{t}$ is a bijective, monotonic continuous mapping. Then the PDF of $\mathbf{Y}$ can be written as

$$p_{\mathbf{Y}}(\mathbf{y}) = |\mathbf{J}| \cdot p_{\mathbf{X}}(\mathbf{t}^{-1}(\mathbf{y}))$$

where

$$\mathbf{J} = \begin{pmatrix} \frac{\partial \mathbf{t}_1^{-1}}{\partial y_1} & \cdots & \frac{\partial \mathbf{t}_1^{-1}}{\partial y_d} \\ \vdots & & \vdots \\ \frac{\partial \mathbf{t}_d^{-1}}{\partial y_1} & \cdots & \frac{\partial \mathbf{t}_d^{-1}}{\partial y_d} \end{pmatrix} \in \mathbb{R}^{d \times d}$$

is the Jacobian of the inverse of $\mathbf{t}$.

Example: Polar coordinates

Given the two representations of vectors in the plane:

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} r \\ \phi \end{pmatrix}, \quad \mathbf{x} = \mathbf{t}^{-1}(\mathbf{y}) = \begin{pmatrix} r \cdot \cos(\phi) \\ r \cdot \sin(\phi) \end{pmatrix}.$$

Then the Jacobian of the transformation is

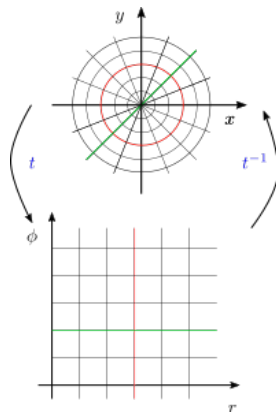
$$\mathbf{J} = \begin{pmatrix} \cos(\phi) & -r \cdot \sin(\phi) \\ \sin(\phi) & r \cdot \cos(\phi) \end{pmatrix},$$

and its determinant is given by

$$\begin{aligned} |\mathbf{J}| &= \cos(\phi) \cdot r \cdot \cos(\phi) - \sin(\phi) \cdot (-r \cdot \sin(\phi)) \\ &= r \cdot (\cos(\phi)^2 + \sin(\phi)^2) = r. \end{aligned}$$

So the PDFs are connected via

$$p_{\mathbf{Y}}(r, \phi) = r \cdot p_{\mathbf{X}}(x, y).$$



In the Inversion Method, we need to compute the CDF of a given PDF.

$$1 = \int_{-\infty}^{\infty} p_{\mathbf{Z}}(\mathbf{z}') d\mathbf{z}' = \int_{z'_d=-\infty}^{\infty} \cdots \int_{z'_1=-\infty}^{\infty} p_{\mathbf{Z}}(z'_1, \dots, z'_d) dz'_1 \dots dz'_d$$

If the different coordinates are independent random variables, we can compute d 1-dimensional PDFs out of a one d -dimensional PDF by integrating all other dimensions and keeping only one dimension free:

$$p_{(k)}(z_k) := \int_{z'_d, \dots, z'_{k+1}, z'_{k-1}, \dots, z'_1 = -\infty}^{\infty} p_{\mathbf{Z}}(z'_1, \dots, z'_d) dz'_1 \dots dz'_{k-1} dz'_{k+1} \dots dz'_d$$

This function is called Marginal Density Function and can also be regarded as a PDF:

$$\int_{z'_k=-\infty}^{\infty} p_{(k)}(z'_k) dz'_k = \int_{z'_d=-\infty}^{\infty} \cdots \int_{z'_1=-\infty}^{\infty} p_{\mathbf{Z}}(z'_1, \dots, z'_d) dz'_1 \dots dz'_d = 1$$

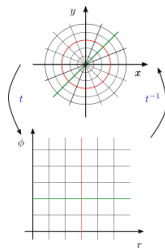
Example: Sampling of a disk

Consider the constant PDF on the unit circle:

$$p_{\mathbf{X}}(x, y) = \frac{1}{\pi}$$

Using the transformation to polar coordinates, we have

$$p_{\mathbf{Y}}(r, \phi) = r \cdot p_{\mathbf{X}}(x, y) = \frac{r}{\pi}$$



Since the coordinates are independent, we can compute the CDF in the following way:

$$p_{(1)}(r) = \int_0^{2\pi} \frac{r}{\pi} d\phi' = 2r$$

$$p_{(2)}(\phi) = \int_0^1 \frac{r'}{\pi} dr' = \frac{1}{2\pi}$$

$$P_{(1)}(r) = \int_0^r 2r' dr' = r^2$$

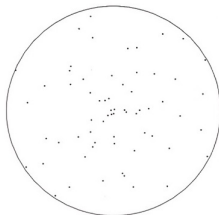
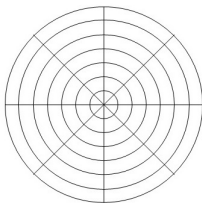
$$P_{(2)}(\phi) = \int_0^\phi \frac{1}{2\pi} d\phi' = \frac{\phi}{2\pi}$$

Let ξ_1, ξ_2 be random samples drawn uniformly from $[0, 1]$. Finally, we get

$$r = P_{(1)}^{-1}(\xi_1) = \sqrt{\xi_1} \quad , \quad \phi = P_{(2)}^{-1}(\xi_2) = 2\pi\xi_2$$

Example: Sampling of a disk

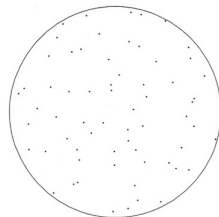
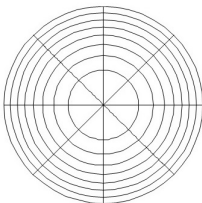
wrong $\neq$ equiareal



$$r = \xi_1$$

$$\theta = 2\pi\xi_2$$

right = equiareal



$$r = \sqrt{\xi_1}$$

$$\theta = 2\pi\xi_2$$



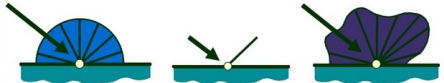
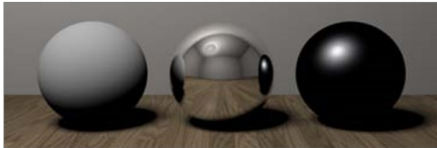
Example: BRDF Sampling



Now, we apply our new knowledge to the Rendering Equation which is defined as

$$L_o(\mathbf{x}, \Omega_o) = L_e(\mathbf{x}, \Omega_o) + \int_{\mathcal{H}_i} f_{BRDF}(\Omega_i, \mathbf{x}, \Omega_o) L_o(\mathbf{h}(\mathbf{x}, \Omega_i), -\Omega_i) \cos(\theta_i) d\omega_i$$

Problem: Typically, the BRDF's $f_{BRDF}(\Omega_i, \mathbf{x}, \Omega_o)$ of the modeled materials are known. However, the illumination $L_o(\mathbf{h}(\mathbf{x}, \Omega_i), -\Omega_i)$ at the surfaces is not known.



Idea: Sample the BRDF (and the cosine) using the Monte-Carlo Estimator

$$L_o(\mathbf{x}, \Omega_o) \approx L_e(\mathbf{x}, \Omega_o) + \sum_{i=1}^n \frac{f_{BRDF}(\hat{\Omega}_i, \mathbf{x}, \Omega_o) \cos(\hat{\theta}_i)}{p(\mathbf{x}, \hat{\Omega}_i)} L_o(\mathbf{h}(\mathbf{x}, \hat{\Omega}_i), -\hat{\Omega}_i)$$

where p is defined over the hemisphere $\mathcal{H}_i$.



Example: Most surfaces reflect diffusely, so try to generate samples for that case:

$$f_{BRDF}(\Omega_i, \mathbf{x}, \Omega_o) \cos(\theta_i) = \frac{\kappa_d \cos(\theta_i)}{\pi}$$

Following the principle of Importance Sampling, we should choose the PDF

$$p(\Omega_i) = c \cdot \cos(\theta_i)$$

This is often called **Cosine Theta Sampling**. Note that we can neglect the dependency on $\mathbf{x}$. Now, we compute the constant c :

$$1 = \int_{\mathcal{H}_i} c \cdot \cos(\theta_i) d\omega_i = \int_{\phi_i=0}^{2\pi} \int_{\theta_i=0}^{\pi/2} c \cdot \cos(\theta_i) \sin(\theta_i) d\theta_i d\phi_i = c \cdot \pi$$

So choose the PDF

$$p(\Omega_i) = \frac{\cos(\theta_i)}{\pi}$$

Equivalently, if we solved the Rendering Equation in spherical coordinates (θ_i, ϕ_i) (which leads to a slightly different Monte-Carlo Estimator!), we would get the PDF

$$p(\theta_i, \phi_i) = \frac{\cos(\theta_i) \sin(\theta_i)}{\pi}$$



From now on, one would compute the CDFs, invert them and then get formulae to sample θ and ϕ . However, we are only interested in the direction that is sampled.

Idea: Uniformly generate points on the unit disk and then project them up to the hemisphere above to get the final directions

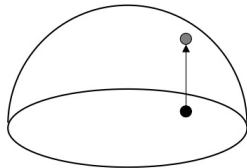
Algorithm Malley's Method

- 1: Generate random samples ξ_1, ξ_2 uniformly from $[0, 1]$
- 2: Compute point on unit disk

$$\begin{aligned} r &= \sqrt{\xi_1} & \phi_i &= 2\pi\xi_2 \\ x &= r \cdot \cos(\phi_i) & y &= r \cdot \sin(\phi_i) \end{aligned}$$

- 3: Project the point up to the hemisphere

$$z = \sqrt{\max\{0, 1 - x^2 - y^2\}}$$



Note: The maximum function is used here to guarantee numerical stability!



Question: But why does Malley's Method work?

Consider the two sampling domains with the PDF

$$\mathbf{x} = \begin{pmatrix} r \\ \phi_i \end{pmatrix} \quad , \quad \mathbf{y} = \begin{pmatrix} \theta_i \\ \phi_i \end{pmatrix} \quad , \quad p_{\mathbf{X}}(\mathbf{x}) = \frac{r}{\pi}$$

Mapping r to the hemisphere leads to

$$r = \sin(\theta_i)$$

Now consider the mapping

$$\mathbf{x} = \mathbf{t}^{-1}(\mathbf{y}) = \begin{pmatrix} \sin(\theta_i) \\ \phi_i \end{pmatrix}$$

Then the determinant of the Jacobian of the transformation is

$$|\mathbf{J}| = \left| \begin{pmatrix} \cos(\theta_i) & 0 \\ 0 & 1 \end{pmatrix} \right| = \cos(\theta_i)$$

So we get exactly the PDF which we want to have:

$$p(\theta_i, \phi_i) = p_{\mathbf{Y}}(\mathbf{y}) = |\mathbf{J}| \cdot p_{\mathbf{X}}(\mathbf{x}) = \frac{\cos(\theta_i) \sin(\theta_i)}{\pi}$$



Malley's Methode delivers cosine-weighted unit-length vectors on the hemisphere parameterized over uniformly distributed values (ξ_1, ξ_2) on the unit-square:

$$\hat{s} = \begin{pmatrix} \sqrt{(\xi_1)} \cos(2\pi\xi_2) \\ \sqrt{(\xi_1)} \sin(2\pi\xi_2) \\ \sqrt{(1 - \xi_1)} \end{pmatrix}$$

The **Elliptic Stretch** bends micro-facet normals towards the geometry normal according to the roughness parameters α_x and α_y :

$$\hat{m} = \text{normalize} \begin{pmatrix} \alpha_x \cdot \hat{s}_x \\ \alpha_y \cdot \hat{s}_y \\ \hat{s}_z \end{pmatrix}$$

Combining the cosine-weighted hemisphere with the elliptic stretch corresponds to the prescribed probability density function⁷ of the GGX-distribution

$$p(\hat{m}) = D(\hat{m}) \cdot (\hat{m} \cdot \hat{n}), \text{ where } D(\hat{m}) = \frac{1}{\pi \alpha_x \alpha_y \left(\frac{\hat{m}_x^2}{\alpha_x^2} + \frac{\hat{m}_y^2}{\alpha_y^2} + \hat{m}_z^2 \right)^2}$$

⁷P.-K. Engstad. *Elliptic Stretch*. <https://engstad.github.io/elliptic-stretch.html>. 2020 (accessed June 2, 2020).



Problem: If the Jacobian derivative matrix $\mathbf{J}$ is non-square, i.e. $\mathcal{V}_X \subset \mathbb{R}^k$ and $\mathcal{V}_Y \subset \mathbb{R}^n$ with $n \neq k$, $|\mathbf{J}|$ is not defined.

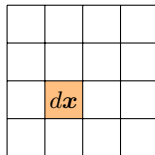
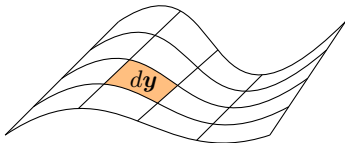
Definition (Manifold)

Let $\mathcal{V} \subset \mathbb{R}^n$ be continuous domain. $\mathcal{V}$ is called k -manifold if it locally resembles $\mathbb{R}^k$ at every point $x \in \mathcal{V}$, for $k \leq n$. I.e. for every $x \in \mathcal{V}$ there is a small region $\mathcal{B}_\varepsilon(x)$ around x that is homeomorphic to $\mathbb{R}^k$.

Examples:

- ▶ $\mathbb{R}^n$ is an n -manifold in $\mathbb{R}^n$.
- ▶ The graph of a function $\{(x, f(x))^t \mid x \in (0, 1)\}$ is a 1-manifold in $\mathbb{R}^2$ for $f : (0, 1) \rightarrow \mathbb{R}$ continuous.
- ▶ The sphere S^2 is a 2-manifold in $\mathbb{R}^3$.

Problem: If the Jacobian derivative matrix $\mathbf{J}$ is non-square, i.e. $\mathcal{V}_X \subset \mathbb{R}^k$ and $\mathcal{V}_Y \subset \mathbb{R}^n$ with $n \neq k$, $|\mathbf{J}|$ is not defined.



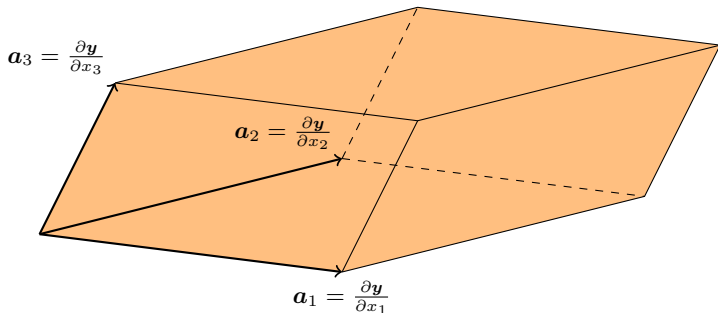
We are only interested in the differential volume *inside* of the the manifold. In the following we consider the parallelepiped that is spanned by the partial derivative vectors $\frac{\partial \mathbf{y}}{\partial x_i}$ and $\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = (\frac{\partial \mathbf{y}}{\partial x_1}, \dots, \frac{\partial \mathbf{y}}{\partial x_k})$.

Require: $k \leq n$ and $\mathcal{V}_Y$ is k -manifold.

Theorem (Volume of k -parallelepiped in $\mathbb{R}^n$)

Let $\mathbf{A} = \{\mathbf{a}_1, \dots, \mathbf{a}_k\} \in \mathbb{R}^{n \times k}$ be a matrix of basis vectors spanning a parallelepiped with $k \leq n$. Then the (k -dimensional) volume of the parallelepiped is

$$\text{vol}_k(\mathbf{A}) = \text{vol}_k(\mathbf{a}_1, \dots, \mathbf{a}_k) = \sqrt{|\mathbf{A}^t \cdot \mathbf{A}|}$$





Proof.

Consider $n = k$ at first:

$$\text{vol}_k(\mathbf{A}) = \sqrt{|\mathbf{A}^t \cdot \mathbf{A}|} = \sqrt{|\mathbf{A}^t| \cdot |\mathbf{A}|} = \sqrt{|\mathbf{A}|^2} = |\mathbf{A}| = \text{vol}(\mathbf{A})$$

For $k < n$ we can write

$$\mathbf{A} = \mathbf{Q} \cdot \mathbf{R},$$

with unitary matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ (i.e. $\mathbf{Q}$ is a rotation matrix),
upper triangular matrix $\mathbf{R} \in \mathbb{R}^{n \times k}$ (i.e. $\mathbf{R} = (\tilde{\mathbf{R}}^t, \mathbf{0})^t$ with $\tilde{\mathbf{R}} \in \mathbb{R}^{k \times k}$).

Observe that we want $\text{vol}_k(\mathbf{A}) = \text{vol}_k(\tilde{\mathbf{R}})$ by definition since $\tilde{\mathbf{R}}$ spans the k -manifold that $\mathbf{A}$ is homomorphic with and $\mathbf{Q}$ does not scale the volume.

$$\begin{aligned} \text{vol}_k(\mathbf{A}) &= \sqrt{|\mathbf{A}^t \cdot \mathbf{A}|} = \sqrt{|(\mathbf{Q}\mathbf{R})^t \cdot \mathbf{Q}\mathbf{R}|} = \sqrt{|\mathbf{R}^t \mathbf{Q}^t \cdot \mathbf{Q}\mathbf{R}|} \\ &= \sqrt{|\mathbf{R}^t \cdot \mathbf{R}|} = \sqrt{|\tilde{\mathbf{R}}^t \cdot \tilde{\mathbf{R}}|} = \text{vol}_k(\tilde{\mathbf{R}}) \end{aligned}$$

□

Theorem (Transformation of Variables on Manifolds)

Let $\mathbf{X}$ be a random variable over the domain $\mathcal{V}_{\mathbf{X}} \subset \mathbb{R}^k$ with probability density function (PDF) $p_{\mathbf{X}}$. Furthermore, let $\mathbf{Y}$ be a derived random variable forming a k -manifold over the domain $\mathcal{V}_{\mathbf{Y}} \subset \mathbb{R}^n$ with $\mathbf{Y} = \mathbf{t}(\mathbf{X})$ where $\mathbf{t}$ is a bijective, monotonic continuous mapping. Then the PDF of $\mathbf{Y}$ can be written as

$$p_{\mathbf{Y}}(\mathbf{y}) = \text{vol}_k \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right)^{-1} \cdot p_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{\left| \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right)^t \left(\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right) \right|}} \cdot p_{\mathbf{X}}(\mathbf{x}),$$

where $\mathbf{x} := \mathbf{t}^{-1}(\mathbf{y})$.



Recap: Malley's Method:

- ▶ Given $\mathbf{u} = (u_1, u_2) \in [0, 1]^2$ uniformly distributed with $p_U(\mathbf{u}) = 1$.
- ▶ Transform $\mathbf{u}$ to a point uniformly distributed on the disk:

$$\begin{aligned} r &= \sqrt{u_1} & \phi &= 2\pi \cdot u_2 \\ x &= r \cdot \cos(\phi) & y &= r \cdot \sin(\phi) \end{aligned}$$

- ▶ Project the point up to the hemisphere

$$z = \sqrt{1 - x^2 - y^2}$$

- ▶ Output $\boldsymbol{\omega} = (x, y, z)^t \in \mathbb{R}^3$

Todo: Show that

$$p_{\Omega}(\boldsymbol{\omega}) = \frac{z}{\pi} \left(= \frac{\cos(\theta)}{\pi} \right)$$



Proof.

$$\omega = \begin{pmatrix} \sqrt{u_1} \cos(2\pi u_2) \\ \sqrt{u_1} \sin(2\pi u_2) \\ \sqrt{1-u_1} \end{pmatrix}$$

$$\frac{\partial \omega}{\partial \mathbf{u}} = \begin{pmatrix} \partial x / \partial u_1 & \partial x / \partial u_2 \\ \partial y / \partial u_1 & \partial y / \partial u_2 \\ \partial z / \partial u_1 & \partial z / \partial u_2 \end{pmatrix} = \begin{pmatrix} 0.5 / \sqrt{u_1} \cdot \cos(2\pi u_2) & -\sqrt{u_1} \cdot 2\pi \cdot \sin(2\pi u_2) \\ 0.5 / \sqrt{u_1} \cdot \sin(2\pi u_2) & \sqrt{u_1} \cdot 2\pi \cdot \cos(2\pi u_2) \\ -0.5 / \sqrt{1-u_1} & 0 \end{pmatrix}$$

$$\begin{aligned} p_{\Omega}(\omega) &= \text{vol}_k \left(\frac{\partial \omega}{\partial \mathbf{u}} \right)^{-1} \cdot p_U(\mathbf{u}) = \frac{1}{\sqrt{\left| \left(\frac{\partial \omega}{\partial \mathbf{u}} \right)^t \cdot \left(\frac{\partial \omega}{\partial \mathbf{u}} \right) \right|}} = \frac{1}{\sqrt{\left| \begin{pmatrix} \frac{0.25}{u_1} + \frac{0.25}{1-u_1} & 0 \\ 0 & u_1 4\pi^2 \end{pmatrix} \right|}} \\ &= \frac{1}{\pi \sqrt{1 + \frac{u}{1-u_1}}} = \frac{1}{\pi \sqrt{\frac{1-u+u}{1-u}}} = \frac{\sqrt{1-u}}{\pi} = \frac{z}{\pi} \end{aligned}$$

□



Using the k -parallelepiped volume version of the *Transformation of Variables* theorem is often quite tedious. The *cylindrical coordinates* actually map uniformly onto the (hemi)sphere. Once we have shown this, you can use the regular *Transformation of Variables* theorem from the lecture to transform your random variables onto the cylindrical coordinates to determine the PDF.

- ▶ Given $\mathbf{u} = (u_1, u_2) \in [0, 1]^2$ uniformly distributed with $p_U(\mathbf{u}) = 1$.
- ▶ Transform $\mathbf{u}$ to a point uniformly distributed on the hemisphere in cylindrical coordinates:

$$z = u_1$$

$$\phi = 2\pi \cdot u_2$$

- ▶ Compute cartesian coordinates:

$$x = \sqrt{1 - z^2} \cdot \cos(\phi)$$

$$y = \sqrt{1 - z^2} \cdot \sin(\phi)$$

- ▶ Output $\boldsymbol{\omega} = (x, y, z)^t \in \mathbb{R}^3$

Todo: Show that

$$p_{\Omega}(\boldsymbol{\omega}) = \frac{1}{2\pi}$$



Proof.

$$\omega = \begin{pmatrix} \sqrt{1-u_1^2} \cdot \cos(2\pi u_2) \\ \sqrt{1-u_1^2} \cdot \sin(2\pi u_2) \\ u_1 \end{pmatrix}$$

$$\frac{\partial \omega}{\partial \mathbf{u}} = \begin{pmatrix} \partial x / \partial u_1 & \partial x / \partial u_2 \\ \partial y / \partial u_1 & \partial y / \partial u_2 \\ \partial z / \partial u_1 & \partial z / \partial u_2 \end{pmatrix} = \begin{pmatrix} -u_1 / \sqrt{1-u_1^2} \cdot \cos(2\pi u_2) & -\sqrt{1-u_1^2} \cdot 2\pi \cdot \sin(2\pi u_2) \\ -u_1 / \sqrt{1-u_1^2} \cdot \sin(2\pi u_2) & \sqrt{1-u_1^2} \cdot 2\pi \cdot \cos(2\pi u_2) \\ 1 & 0 \end{pmatrix}$$

$$\begin{aligned} p_{\Omega}(\omega) &= \text{vol}_k \left(\frac{\partial \omega}{\partial \mathbf{u}} \right)^{-1} \cdot p_U(\mathbf{u}) = \frac{1}{\sqrt{\left| \left(\frac{\partial \omega}{\partial \mathbf{u}} \right)^t \cdot \left(\frac{\partial \omega}{\partial \mathbf{u}} \right) \right|}} = \frac{1}{\sqrt{\left| \begin{pmatrix} u_1^2 / (1-u_1^2) + 1 & 0 \\ 0 & (1-u_1^2) 4\pi^2 \end{pmatrix} \right|}} \\ &= \frac{1}{2\pi \sqrt{u_1^2 + (1-u_1^2)}} = \frac{1}{2\pi} \end{aligned}$$

□

Proof.

Knowing that $p_{\Omega}(\omega) = p_C(c)$ we can show that Malley's Method works much quicker. The cylindrical coordinates $c = (z, \phi)^t$ using Malley's Method are:

$$c = (z, \phi)^t = (\sqrt{1 - u_1}, 2\pi \cdot u_2)^t$$

$$p_C(c) = \left| \frac{\partial c}{\partial u} \right|^{-1} \cdot p_U(u) = \left| \begin{pmatrix} \frac{-0.5}{\sqrt{1-u_1}} & 0 \\ 0 & 2\pi \end{pmatrix} \right|^{-1} = \left(\frac{\pi}{\sqrt{1-u_1}} \right)^{-1} = \frac{z}{\pi}$$

□



Question: How to deal with more difficult distributions?

Sampling Importance Resampling⁸ (simpler, importance resampling) is a common method in computational statistics for generating samples from difficult distributions. Given a target distribution q it first generates a set of “proposal” samples x_i from a source distribution p , weight these samples by $w(x_i) = \frac{q(x_i)}{p(x_i)}$, then resample these samples by drawing samples from them with probability proportional to their weights.

Algorithm Importance Resampling

Require: M proposals, $M \geq 1$

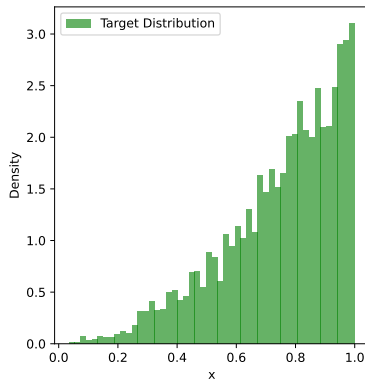
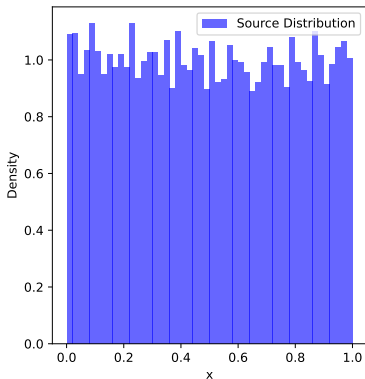
Require: N samples, $N \leq M$

- 1: Generate M proposals from the source distribution p , $\{x_1, \dots, x_M\}$.
 - 2: **for** $j = 1$ to M **do**
 - 3: Compute a weight for each proposal, $w(x_j)$.
 - 4: **end for**
 - 5: Draw N samples, $\{y_1, \dots, y_N\}$, with replacement from the proposals with probability proportional to their weights
-

⁸D. B. Rubin. “Comment: a noniterative sampling/importance resampling alternative to the data augmentation algorithm for creating a few imputations when fractions of missing information are modest: the SIR algorithm”. In: *Journal of the American Statistical Association* 82.398 (1987), pp. 542–543.



Example:



Target distribution $1/3x^2$ generated from a uniform distribution with $M = 10000$ and $N = 8000$.



Definition (RIS Estimator)

Given a target distribution q , a source distribution p and weights $w(x_i) = \frac{q(x_i)}{p(x_i)}$, the **Resampled Importance Sampling Monte Carlo Estimator** (RIS Estimator) is given by

$$\hat{I}_{RIS} = \frac{1}{N} \sum_{i=1}^N \frac{f(y_i)}{\hat{q}(y_i)} \cdot \frac{1}{M} \sum_{j=1}^M w(x_j),$$

For the RIS estimate to be unbiased $M, N \geq 0$ and the sampling density $\hat{q}$ and the proposal density p must be greater than zero everywhere that f is non-zero.

Note, that when $M = N$, the RIS estimate does not reduce to standard Monte Carlo integration, since the effective sample size is smaller than N , because when performing the Sampling Importance Resampling, there is some chance of choosing proposals multiple times. Solutions to this problem are

- ▶ Design a resampling procedure that uses sampling without replacement. Unfortunately, it is difficult to derive an unbiased estimator using this approach.
- ▶ Stratify the proposals. If each proposal belongs to a single stratum, and a single sample is drawn from each stratum then no sample will be replicated.



Unbiasedness of the RIS Estimator⁹ First, recall that

$$I = E_q \left[\frac{f(x)}{q(x)} \right] = \int \frac{f(x)}{q(x)} q(x) dx = \int f(x) \underbrace{\frac{q(x)}{p(x)}}_{\text{importance weight}} p(x) = E_p \left[\frac{f(x)}{p(x)} \right]$$

both deliver unbiased Monte Carlo estimators for the Integral of f .
For the expectation value of the RIS estimator we get

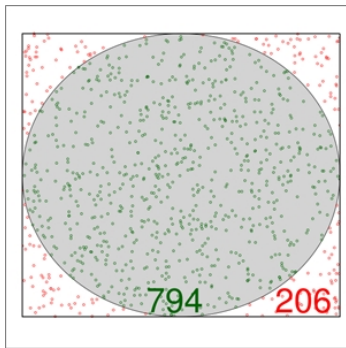
$$\begin{aligned} \mathbb{E}[I_{RIS}] &= E \left[\frac{1}{N} \sum_{i=1}^N \frac{f(y_i)}{\hat{q}(y_i)} \cdot \frac{1}{M} \underbrace{\sum_{j=1}^M w(x_j)}_{=:W} \right] = \frac{1}{N} \frac{W}{M} \sum_{i=1}^N E \left[\frac{f(y_i)}{\hat{q}(y_i)} \right] \\ &= \frac{1}{N} \frac{W}{M} \sum_{i=1}^N \sum_{j=1}^M \frac{f(y_i)}{q(y_i)} \underbrace{P(y_i = x_j)}_{=:w(x_j)/W} = \frac{1}{N} \sum_{i=1}^N \frac{f(x_j)}{p(y_i)} \end{aligned}$$

⁹J. F. Talbot. *Importance resampling for global illumination*. Brigham Young University, 2005.



An alternative to sample from difficult distributions is **rejection sampling**

Motivation: determination of π



$$\frac{\pi}{4} = 0.7854\dots \approx \frac{794}{794+206} = 0.794$$

Efficiency: $\frac{\pi}{4}$



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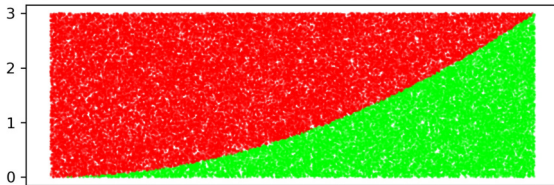


Rejection Sampling

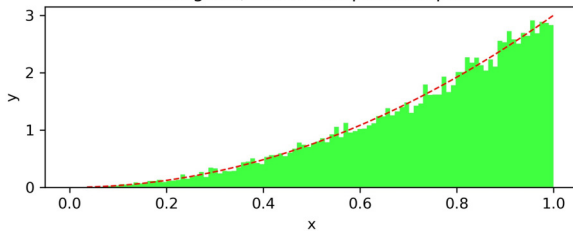


Example:

Rejection sampling of $p(x) = 3 * x^2$, $g(x) = 3$, 50000 samples



Histogram, 16600 samples accepted





Acceptance probability: ξ uniform random sample from $[0, 1]$

$$\begin{aligned} Pr\left(\xi \leq \frac{p(x)}{Mq(x)}\right) &= E_q\left[Pr\left(\xi \leq \frac{p(x)}{Mq(x)} \middle| x\right)\right] = E_q\left[\frac{p(x)}{Mq(x)}\right] \\ &= \int \frac{p(z)}{Mq(z)} q(z) dz = \frac{1}{M} \int p(z) dz = \frac{1}{M} \end{aligned}$$

Number of samples required: To obtain an accepted sample follows a geometric distribution with probability $\frac{1}{M}$ which has mean M . Thus M is the number of iterations needed to get one accepted sample.

Hint: Use sharp upper bound $Mq(x)$ of $p(x)$ to improve the efficiency of the algorithm.

Drawbacks:

- ▶ In high dimensions the algorithm gets inefficient and impractical (curse of dimensionality).
- ▶ In high dimensions, it is necessary to use different approaches, e.g. Metropolis sampling

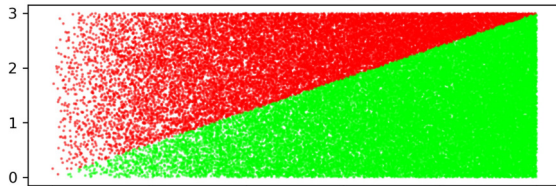


Rejection Sampling

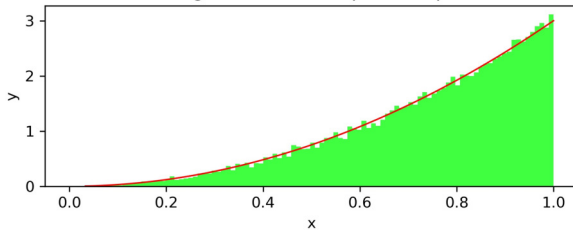


Example:

Rejection sampling for $p(x) = 3 \cdot x^2$ using $q(x) = 3x$, 50000 samples



Histogram, 33308 samples accepted





Let $T = (t_{ij})$ be the transition matrix of an **irreducible Markov chain**¹⁰ with states $0, 1, \dots, M$. If $X(t)$ denotes the state occupied by the process at time t , we have

$$T(X(t+1) = x_j | (X(t) = x_i) = t_{ij}.$$

Let $\pi = (\pi_0, \pi_1, \dots, \pi_M)$ be a probability distribution with $\pi_i > 0 \quad \forall i$. If we choose T such that π is its unique **stationary distribution**, i.e. $\pi = \pi T$, and f be real-valued function on the states, then we can approximate its expectation value with respect to π by simulating the corresponding Markov chain for times $t = 1, \dots, N$ generating the sample values $X(1) = x_1, \dots, X(N) = x_N$

$$E_{\pi}(f) = \sum_{i=0}^M f(i)\pi_i \approx \frac{1}{N} \sum_{t=1}^N f(X(t))$$

¹⁰In an irreducible Markov chain, it is possible to reach any state from any other state in a finite number of steps, meaning that the states are all interconnected. In other words, given any two states i and j in an irreducible Markov chain, there exists a sequence of transitions (steps) that can take you from state i to state j , regardless of the initial state. This property ensures that no state is completely isolated or disconnected from the rest of the chain.



How to construct the transition matrix P ?

- ▶ Let p be the propability density on our state space we want to sample from and $t(x_j|x_i) = \forall x_i, x_j$ be the transition probabilities of an irreducible Markov chain determined by the transition matrix $T = t(x_j|x_i)_{(ij)} = t_{ij}$
- ▶ Hastings¹¹ constructed the transition matrix such that its unique stationary distribution $\pi = \pi T$ is proportional to p :
 - ▶ T is required to satisfy, for all pairs of states x, y the reversibility condition

$$p(x_i)t(x_j|x_i) = p(x_j)t(x_i|x_j)$$

- ▶ It is further assumed that

$$t(x_j|x_i) = \begin{cases} t(x_j|x)a(x_j|x_i) & \text{if } (x_i \neq x_j) \\ 1 - \int p(x_j|x_i)dy & x_j = x_i \end{cases}$$

where the acceptance probabilities $a(x_j|x_i)$ are given by

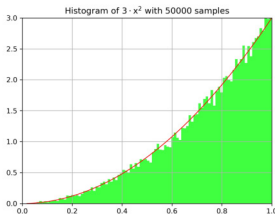
$$a(x_j|x_i) = \min \left(1, \frac{p(x_j)t(x_i|x_j)}{p(x_i)t(x_j|x_i)} \right)$$

¹¹W. K. Hastings. "Monte Carlo sampling methods using Markov chains and their applications". In: *Biometrika* 57.1 (1970), pp. 97–109.



Algorithm Basic Metropolis Sampling algorithm

```
1:  $x = x_0$  ▷ choose starting point
2: for  $i=1$  to  $n$  do
3:    $x' = \text{mutate}(x)$ 
4:    $a = \text{accept}(x, x')$  ▷ calculate acceptance probability for  $x'$ 
5:   if  $\text{random}() < a$  then
6:      $x = x'$  ▷ accept mutation
7:   end if
8:    $\text{record}(x)$  ▷ add sample
9: end for
```





Remarks:

- For a symmetric transition probability, i.e. $t(x_j|x_i) = t(x_i|x_j)$ the acceptance probability simplifies to

$$a(x_j|x_i) = \min \left(1, \frac{p(x_j)}{p(x_i)} \right)$$

- Interpretation: if $p(x_j) \geq p(x_i)$ the new sample x_j is accepted in any case. Otherwise, if $p(x_j) < p(x_i)$ the process jumps into state x_j with probability $\frac{p(x_j)}{p(x_i)}$, only. Intuitively, more samples are generated in areas of the state space where the probability density is higher and less samples where the probability is smaller.
- Note, the user has to ensure that the transition probability matrix T guarantees an irreducible Markov chain, i.e. that each state of the state space is visited while running the Markov chain.



```
function MAKEHISTOGRAM( $F$ ,  $histogram$ ,  $mutations$ )  
   $w, h \leftarrow \text{shape}(F)$   
   $x_0 \leftarrow \text{randomInteger}(0, w - 1)$   
   $x_1 \leftarrow \text{randomInteger}(0, h - 1)$   
   $F_x \leftarrow F[x_0, x_1]$   
   $T_{xy} \leftarrow 1.0/(w * h)$   
   $T_{yx} \leftarrow 1.0/(w * h)$   
  for  $i \leftarrow 0$  to  $mutations$  do  
     $y_0 \leftarrow \text{randomInteger}(0, w - 1)$   
     $y_1 \leftarrow \text{randomInteger}(0, h - 1)$   
     $F_y \leftarrow F[y_0, y_1]$   
     $A_{xy} \leftarrow \min(1, (F_y * T_{xy})/(F_x * T_{yx}))$   
    if  $\text{randomReal}(0.0, 1.0) < A_{xy}$  then  
       $x_0 \leftarrow y_0$   
       $x_1 \leftarrow y_1$   
       $F_x \leftarrow F_y$   
    end if  
     $histogram[x_0, x_1] \leftarrow histogram[x_0, x_1] + 1$   
  end for  
end function
```



Bottom: original image, images at top and middle generated with 8 spp and 256 spp.

¹²D. Cline and P. Egbert. "A practical introduction to metropolis light transport". In: *Brigham Young University* (2005).



Comments:

- ▶ Usually, P is constructed so that it depends only on the ratios $\frac{p_j}{p_i}$, so we need only know the π_i up to a constant of proportionality.
- ▶ We shall apply these methods to distributions defined mathematically on an infinite sample space although, when we actually simulate the Markov chains on a digital computer, we will have a large but finite number of states.
- ▶ The Metropolis–Hasting-algorithm do not suffer from the curse of dimensionality problem like e.g. rejection based methods.
- ▶ If the number of dimensions is high it is one of the available solutions
- ▶ The samples generated by the algorithm are correlated. To avoid this, only each n th ($n > 1$ fixed) sample generated during the random walk is actually used. The others are thrown away
- ▶ The initial samples may follow a very different distribution, especially if the starting point is in a region of low density. Therefore, it is reasonable to skip e.g. the first 1000 samples.



- ▶ Given the integral

$$\int_{\Omega} f(x)g(x)dx$$

- ▶ Standard Monte Carlo estimator:

$$\int_{\Omega} f(x)g(x)dx \approx \frac{1}{N} \sum_{i=1}^N \frac{f(x_i)g(x_i)}{p(x_i)},$$

where $x_i \propto p(x)$, an arbitrary pdf

- ▶ Metropolis gives $x_1, \dots, x_N, x_i \propto f_{pdf}(x)$

$$\int_{\Omega} f(x)g(x)dx \approx \left[\frac{1}{N} \sum_{i=1}^N \frac{g(x_i)f(x_i)}{f(x_i)/I(f)} \right] = \left[\frac{1}{N} \sum_{i=1}^N g(x_i) \right] \cdot I(f),$$

where $I(f) = \int_{\Omega} f(x)dx$.



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